

growth of around 170 per cent. A total of \$376 million was invested in 2019 in EVs. These figures were \$147 million in 2018.

Is the Tesla story relevant for India?

Tesla, no doubt, is the giant of four-wheel electric cars in the world. The company's market cap has already jumped by a humongous \$500 billion in 2020. It sells the base model of Tesla 3 north of 35,000 dollars. Several

e-commerce car portals estimate that the entry price for the same model in India (once Tesla is in India in 2021) would be around six million rupees, which is almost double its price in the USA. The high cost, in itself, might be a big deterrent for Tesla becoming as good an EV company in India as it is in the United States.

"Tesla arguably created the EV ecosystem for the western world. That is what I look at Tesla as. It also is noted that Tesla has been supported significantly during its long loss-making period by the government of the United States of America. Elon Musk, himself had a very good capability of raising funds. That is how the company managed to sustain 14-15 years of heavy loss making," explains Rajat Verma, founder and CEO, Lohum Cleantech.

While Elon Musk joined Tesla in 2004, the EV company was founded in 2003 by Martin Eberhard and Marc Tarpennig. Musk had initially invested \$6.3 million in Tesla stocks during a Series A round of investment. It was in 2019 (around 16 years after Tesla was founded) that Tesla had posted figures claiming first annual profit.

"That is something that an EV maker based out of India may not be able to afford. But it is also very important to understand that we have opportunities in various other categories of EVs," adds Verma.

Tesla, on an average, sells around 200,000 EVs in the USA annually. The same translates to around twelve gigawatts of energy capacity that the company delivers annually. In comparison, there are about 1.5 million



E-rickshaws operating on the Indian roads, translating to around six gigawatts of energy capacity.

Rajat adds, "If we can even convert around five million vehicles into electric ones, we are talking about ten gigawatts in energy capacity. India has a lot of other variants in terms of EVs than just passenger cars. We have the ability to create a very large energy capacity in the market."

As per Statista, the number of two-wheelers sold in India during 2019 was around 21.8 million, three wheelers 0.7 million, and commercial vehicles around 1.0 million. The number of passenger cars sold in the same year was 3.4 million.

"Four-wheeler EV cars will not probably solve the ecosystem problems ranging from raw materials, battery recycling, motors, cell manufacturing, charging infrastructure, financing, and insurance. To do that I think we need an EV car company to lead the race, but two-wheeler and three-wheeler probably will come out as the surprise categories in India," says Verma.

EV market, in India, as explained by Alwala Devender Reddy, CEO, E-ride E-mobility Company, is being created more by two-wheelers and three-wheelers than four-wheelers. The three-wheelers market is further divided into logistics and passenger categories. He explains that while e-commerce companies are adopting two-wheelers for deliveries, three-wheelers are mostly being used for last-mile connectivity.

"Data proves that the number of people who have used shared electric

three-wheelers are more than the people who own their personal vehicles in India. A person sitting in an electric bus or electric metro will not understand the benefits offered by an EV unless he experiences owning one himself. In India, the easier way to own an electric vehicle would either be through a two- or a three-wheeler," as per Reddy.

Ranjita Ravi, director, Orxa Energies,

notes, "What I like about Tesla is the way they started and proved what electric vehicles can actually do. Tesla proved that electric vehicles could give high performance, be fast, be good-looking and very aspirational at the same time as compared to a normal internal combustion engine (ICE) powered car. And now they have proved that EVs are a good fit for the economical category too."

Is focus on different EV categories a roadblock?

One of the prime reasons behind Tesla becoming a success story was the company's capability to bring the entire EV ecosystem together. As of today, most of the parts and raw materials Tesla uses to manufacture cars in the USA are sourced from within the country, and for the ones that Tesla relies on outside sources, the process is so smooth that there are no chances of bottlenecks or roadblocks. It is like Tesla had a clear aim of being a specialist in the four-wheeler EVs from the beginning.

"Tesla has become what it is today because its aim from the very start was crystal clear. We, in India, already have EV categories that are on the brink of inflection. The three-wheeler and two-wheeler EV categories have already shown a lot of promise. It does seem that economic viability in these two categories is already there," points out Verma.

He continues, "India should focus on two-wheeler and three-wheeler EV categories for creating a robust ecosystem for electric vehicles. These two have the potential to help create